HOMEWORK 1.1 · UNIT 1 · NC.8.NS.1

Real Number System Review

Aug 5 / Aug 6 · Targets: NS.1-3, NS.1-1, NS.1-4, NS.1-2, NS.1-6, NS.1-7, NS.1-8, NS.1-9

Name _____ Date _____ Block _____

Remember: Rational = can be written as a fraction $\frac{a}{b}$ of two integers. Its decimal **stops** (terminating) or **repeats**. **Irrational** = the decimal **never stops and never repeats**, like π or $\sqrt{2}$. Sets inside sets: **Natural** {1, 2, 3, ...} ⊂ **Whole** {0, 1, 2, ...} ⊂ **Integers** {..., −2, −1, 0, 1, 2, ...} ⊂ **Rational**. Perfect squares: 1, 4, 9, 16, 25, 36, 49, 64, 81, 100.

SCORE 3.0 Approaching · the basics

Targets: NS.1-3, NS.1-1, NS.1-4, NS.1-2

1 Circle one: $\sqrt{25}$ NS.1-3

Rational Irrational

Why? _____

2 Circle one: $\sqrt{7}$ NS.1-3

Rational Irrational

Why? _____

3 $\frac{2}{3}$ = 0.666... as a decimal. What kind of decimal is it? NS.1-1

(A) terminating
(B) repeating
(C) never stops, never repeats

4 0 belongs to which sets? NS.1-4

Hint: Natural numbers start at 1.

- ☐ Natural
- ☐ Whole
- ☐ Integer
- ☐ Rational
- ☐ Irrational

Select all that apply.

5 Every natural number is also a whole number. NS.1-2

True False

SCORE 4.0 Proficient · on target

Targets: NS.1-6, NS.1-7

6 **Leo** made a mistake. Circle it. Explain it. Fix it. NS.1-6

LEO'S WORK
$\sqrt{36}$ is irrational
because it has a square root sign.

7 Which number is rational but **not** an integer? NS.1-7

Ⓐ -8 Ⓑ $\sqrt{3}$ Ⓒ $\frac{5}{4}$ Ⓓ π

8 Write one number that is an integer but **not** a whole number. NS.1-7

Hint: Think below zero.

SCORE 5.0 Mastery · go deeper

Targets: NS.1-8, NS.1-9

9 Explain where **0** goes in the real number system diagram, and why it does not go in the Natural box. NS.1-8

0 goes in the _____ box because _____ .

10 Kai says, "$\frac{\pi}{2}$ is rational because it is a fraction." Do you agree? Explain. NS.1-9

I _____ because _____ .